

1. Prove that for all ints p , p is even if and only if $p^3 + 3$ is odd.

Case 1: p is even	Case 2: p is odd
$p = 2k$	$p = 2k + 1$
$(2k)^3 + 3 = 8k^3 + 3$	$(2k + 1)^3 + 3 = 8k^3 + 12k^2 + 6k + 4$
- $8(k^3)$ will always be even - Any even integer whose value is added by 3 will always be odd. ($+ 3$)	- $2(4k^3 + 6k^2 + 3k + 2)$ will always be even because it is divisible by 2
Therefore, if p is even integer, then $p^3 + 3$ will always result in being odd.	Therefore, if p is odd integer, then $p^3 + 3$ will always result in being even.

2. I am choosing proof by contradiction because firstly we used this technique when dealing with irrationals/rationals in class and secondly it is hard to set up cases where x can be irrational or rational like one could with even/odd. Instead, it is better to just prove something is rational when it is assumed that it is not.

1. Assume that x is irrational and $3x$ is rational. ($p \rightarrow \neg q \Rightarrow \neg p$)	—
2. Since $3x$ is rational, we can represent it with (p/q) where p and q are integers, q doesn't equal 0, and p and q have no common factors.	$3x = p/q$
3. Solve for x .	$x = p/3q$
4. Since p is rational and $3q$ is also rational, then the division between the two integers is also rational. This is a contradiction in the proof because it is assumed that x is irrational.	$x = p/3q$ RATIONAL
5. The assumption that $3x$ is rational is false. Therefore, through proof by contradiction, it is determined that if x is irrational, then $3x$ must also be irrational.	

3. I chose proof by contrapositive because of the 'or' statement, as it is difficult to work with an 'or' when you are trying to prove something.

1. Assume $\neg(m \geq 8 \vee n \geq 10)$ in order to prove $\neg(mn > 63)$	—
2. De Morgan's Law	$m < 8 \wedge n < 10$

3. Since m and n are integers, this implies...	$m \leq 7 \wedge n \leq 9$
4. So, $mn \leq 63$, which is the exact opposite (not) of the original $p \rightarrow q$ statement in the problem's $p \rightarrow q$.	$mn \leq 63$
5. This contrapositive statement is proven as the validity checks out from the product of 7 and 9 being smaller than 63. Therefore, by proof of contrapositive, "if $mn > 63$, then $m \geq 8$ or $n \geq 10$ " must be true.	$mn \leq 63 \rightarrow m \leq 7 \wedge n \leq 9$

4. x and y are rational, z is irrational. Prove $xy + z$ is irrational

You can possibly use a direct proof for this statement, but you would have to make some assumptions that may not be known entirely. Examples like two rational numbers always multiply into a rational number and adding a rational and irrational number always results in an irrational number. I wrote my proof indirectly and through two contradictions to confirm each of the two assumptions.

Assume xy is irrational and x and y are rational.

1. Convert x and y into ratios of integers that do not equal 0 since they are rational numbers.	$x = a/b$ $y = c/d$
2. Solve for xy .	$xy = ac/bd$
3. CONTRADICTION , as xy was assumed to be irrational but proven to be rational in proof by contradiction. Therefore, if x and y are rational, then xy must also be rational.	—

Assume z is irrational, xy is rational (proven above), and $xy + z$ is rational

1. Set up an equation where $a/b = xy + z$, with a and b being positive nonzero integers representing the resulting value of $xy + z$.	$a/b = xy + z$
2. Isolate/solve for z .	$z = b/a - xy$
3. CONTRADICTION! Since b/a and xy are assumed/proven to be rational, then z also must be rational. But we assumed that z is irrational. Therefore by proof of contradiction, $xy + z$ (where x, y are rational and z is irrational) must be irrational.	—

5. I used direct proof to show that one month must have three meetings because it's very straightforward to show and doesn't require any theorems or extended information.

- Let j represent the amount of meetings per month consistent with every month of the year.
- 26 represents the amount of meetings total in the year, which consists of 12 months.
- Calculate $26/j$, which represents the number of months needed to accommodate how many meetings per month there are. Let $j = 2$ for the worst case scenario if a month doesn't have to have three meetings.
- $26/2 = 13$ months. $10(2) + 2(3) = 26$ or $11(2) + 1(4) = 26$.
- Since there are only 12 months in a year, two meetings per month is not frequent enough to include all the meetings in the year.
- Therefore, at least one month must have at least three meetings in it for all 26 meetings to fall into the year.

6. I used a proof by contrapositive because of the 'or' statement. Using De Morgan's law, I can get rid of the 'or' and replace it with a simpler 'and'.

1. Proving by contrapositive sets up a proof like this!	$\neg(p \geq 5 \vee p \leq -3) \rightarrow \neg((p-1)^2 \geq 16)$
2. By De Morgan's law and flipping inequalities, it converts into this.	$p < 5 \wedge p > -3 \rightarrow (p-1)^2 < 16$
3. Square root the right side of the statement, allowing for an absolute value inequality.	$ p - 1 < 4$
4. Convert absolute value into double sided inequality and solve for p .	$-4 < p - 1 < 4 = -3 < p < 5$
5. Contrapositive is solved and proven. Therefore, if $p \geq 5$ or $p \leq -3$, then $(p-1)^2 \geq 16$ must be true.	$p < 5 \wedge p > -3 = -3 < p < 5$

$\neg \forall \exists \Rightarrow \supset \rightarrow \Leftrightarrow \equiv \leftrightarrow \mathbb{D} \wedge \vee \parallel + \oplus \nabla \neq \top \blacksquare \perp := \vdash \models$

7. My proof is non-constructive because I found some sort of element combination that matches the statement, but I used a different technique in showing that it exists. I used proof by contradiction to show that the sum can be rational and contradict the assumption that it must be irrational.

1. Assume that x and y are irrational and their sum (z) is also irrational.	$x + y$ $z = x + y.$
2. Let y equal the negative value of x , as a negative irrational is still irrational.	$y = -x$
3. Plug back into the sum equation.	$x - x = z$
4. CONTRADICTION! Z 's sum equals 0, which is a rational number. This contradicts the assumption that z (the sum) is irrational. Therefore, by a non-constructive proof of contradiction, there exists irrational numbers x and y whose sum is rational.	$0 = z$